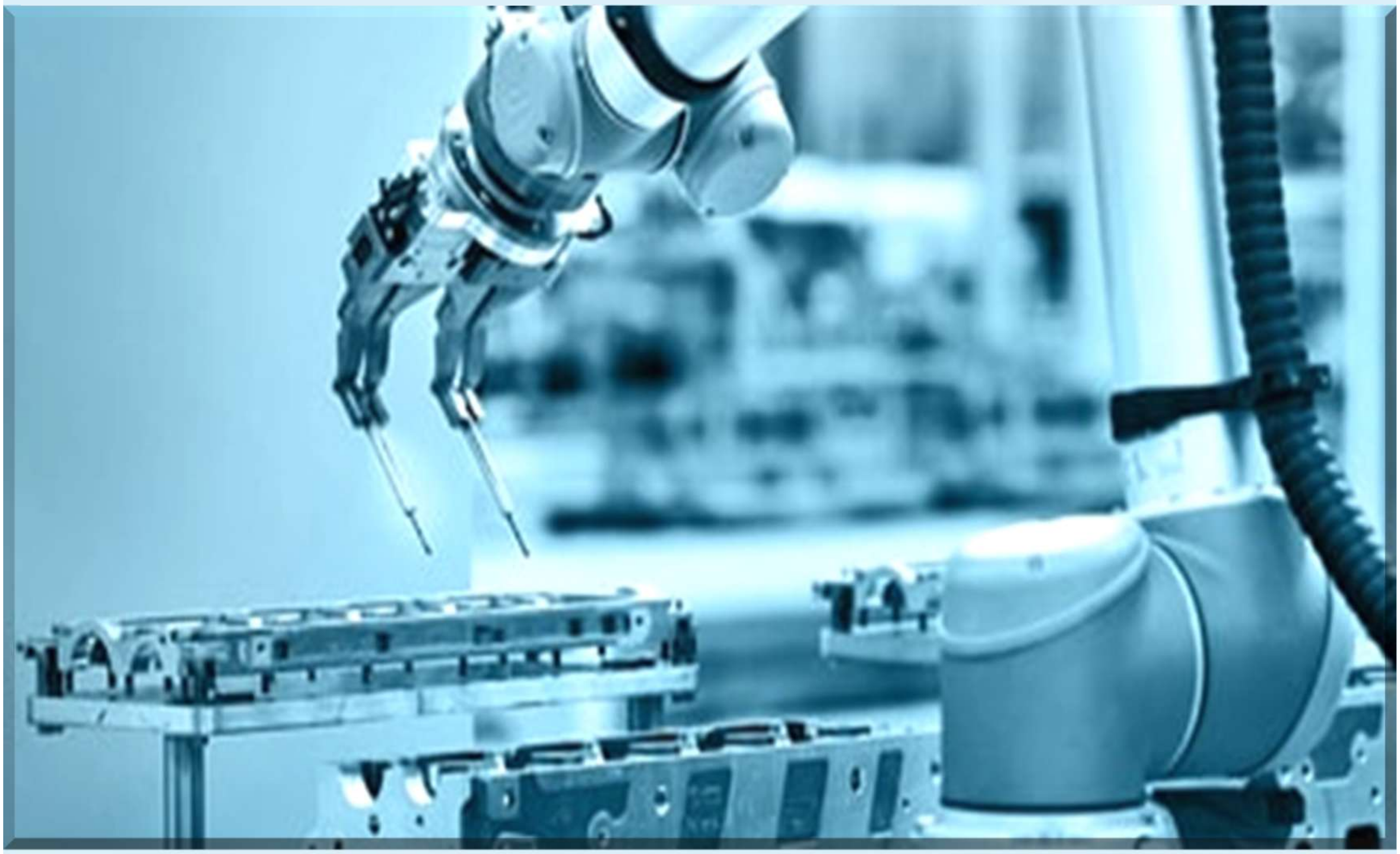
Portfolio



Sayma Sultana Jhara

M.Sc. in Mechanical Engineering

B.Sc. in Mechatronics Engineering

About Me

I am **Sayma**, a **British Council Women in STEM Scholar** with a **Master's in Mechanical Engineering** from *Anglia Ruskin University (UK)* and a *B.Sc. in Mechatronics Engineering*. My research focuses on **robotics,** intelligent automation**,** and **mechatronic-based CAD design**, with several **journal publications**, including work featured in **Springer Nature** on sensors and automation technologies. I am passionate about developing smart and sustainable robotic systems that integrate machine learning and computer vision to address real-world challenges.

Research and Professional Experience

Teaching Assistant · Anglia Ruskin University, UK · Nov – Dec 2025

Work Shadow· Cubic Transport · London, 3 days (July 2025)

Research Assistant ·WUB (Aug 2023-Present)

Trainee QA Engineer (Capture Team), Dhaka · Vcube soft and tech (Sister Concern of Insidemaps, USA) · (Sep 2021 - Jul 2023)

Technical Skills

Language: Python, C, C++, HTML

Software: VS Code, Notepad++, Micro wind, PLC Ladder, MATLAB, Ansys, Autodesk Inventor, ROS2, Gazebo, Proteus, Tinkercad

Technologies/Framework: Linux, GitHub, PyTorch, TensorFlow, OpenCV

Microcontroller: Raspberry Pi, Arduino

Machine Learning & AI: Supervised/Unsupervised Learning, Deep Learning, Computer Vision

Academic Qualification

M.Sc. in Mechanical Engineering
Anglia Ruskin University
UK (2024-2025)

Summer School on Automation and Simulation
RWTH Aachen University
Germany (July 2025)

B.Sc. in Mechatronics Engineering
World University of Bangladesh
Bangladesh (2017-2023)

Contact Information

+447770148283

saymamte@gmail.com

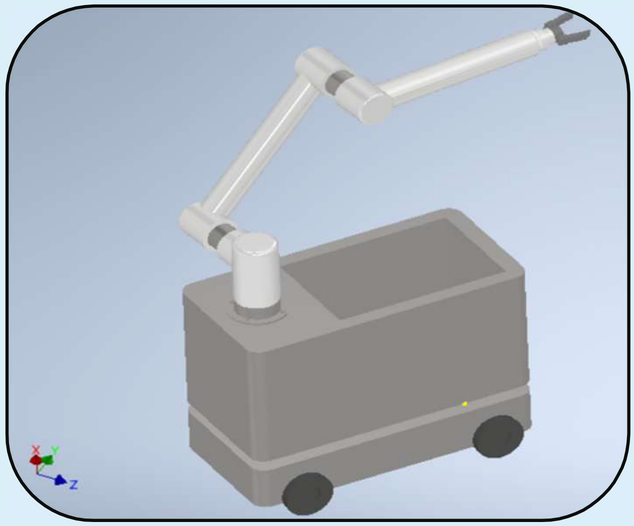
LinkedIn: Sayma Sultana

Google Scholar: Sayma Sultana Jhara

GitHub: Sayma

Contents

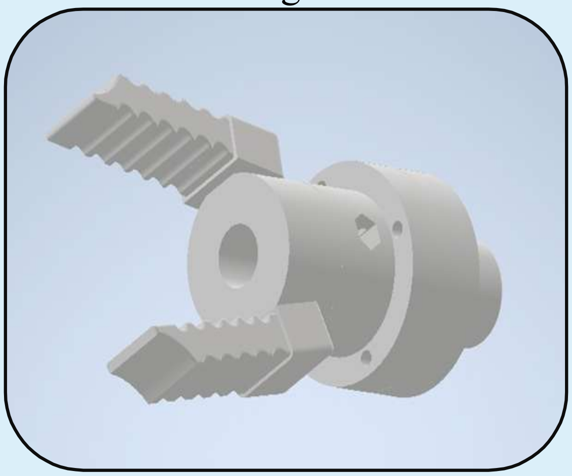
Dissertation Project (Graduate)



Major Project:" *Design and Simulation of a Rule-Based 3-DOF Collaborative Robot for Expiry Food Detection and Waste Reduction in UK Supermarkets"*

Project Place and Duration: Anglia Ruskin University, Jun 2025 – Sept 2025

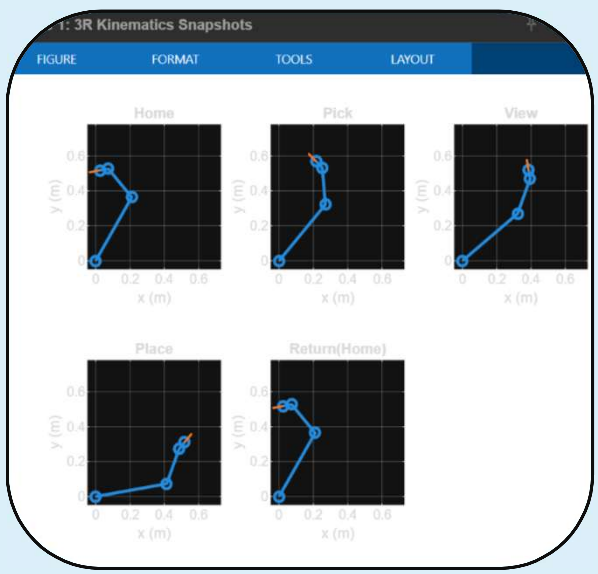
Context: UK supermarkets face growing challenges in reducing food waste due to inefficient monitoring of expiry dates and manual handling.

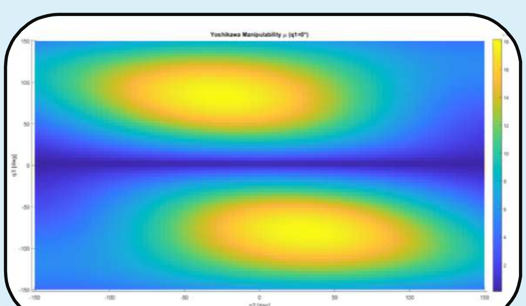


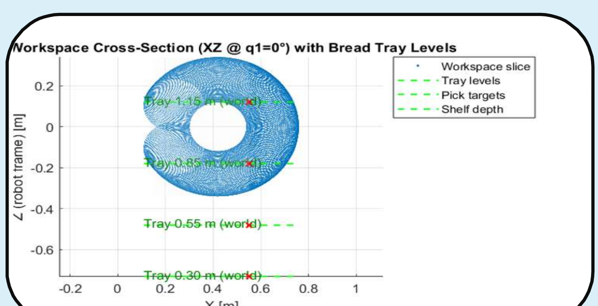
Description: Involved the design and simulation of a rule-based **3-DOF cobot for expired food detection** and waste reduction in UK supermarkets. Using simulation tools, the robot's motion, task execution, and control strategies were modelled and analysed to ensure efficient and safe operation in a collaborative environment.

The study demonstrates the potential of simulation-driven robotic solutions for optimising supermarket operations and minimising food waste.

Pictures show the design of the end effector for **perishable food** items, the 3R kinematics of the Cobot. The other two pictures show the Yoshikawa manipulability index distribution across q2 and q3, as well as the vertical reach of the bot and its compatibility with various bread tray levels.

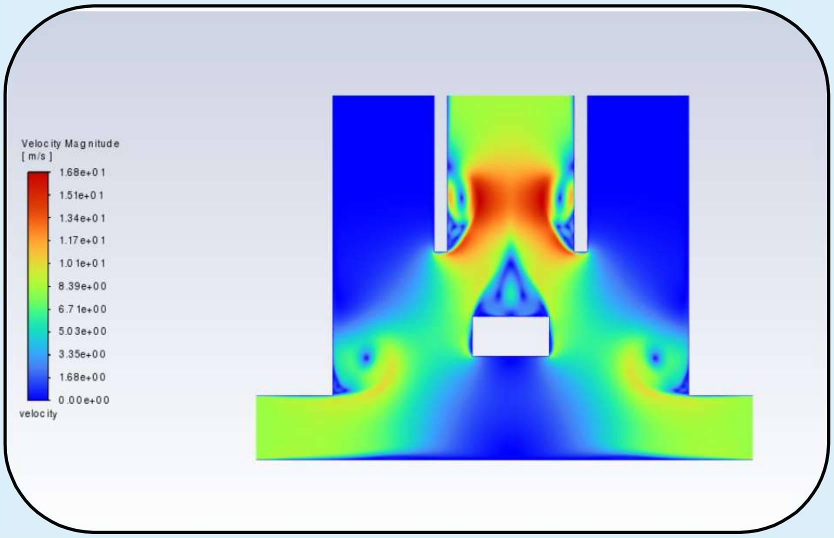






Additional Projects (Graduate)

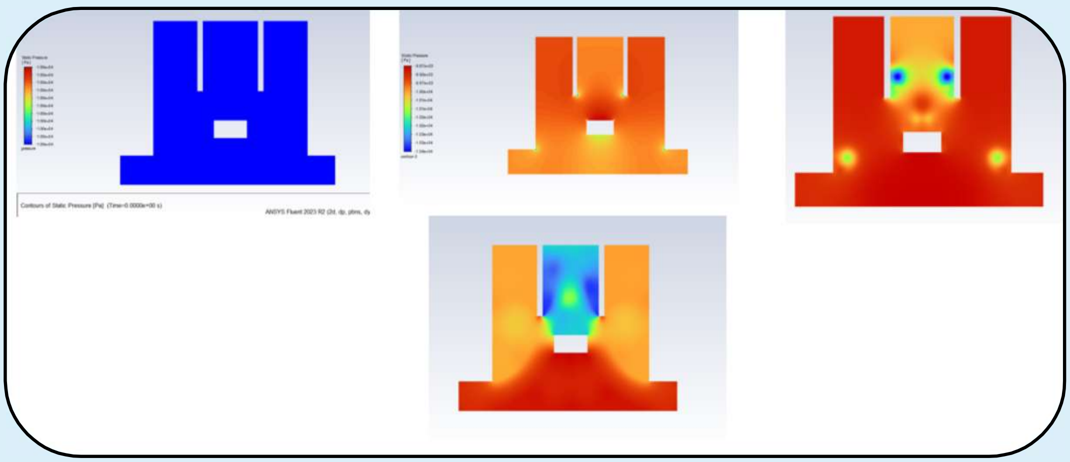
Project: *"CFD project on a 2D unsteady flow model for a suction robot"*

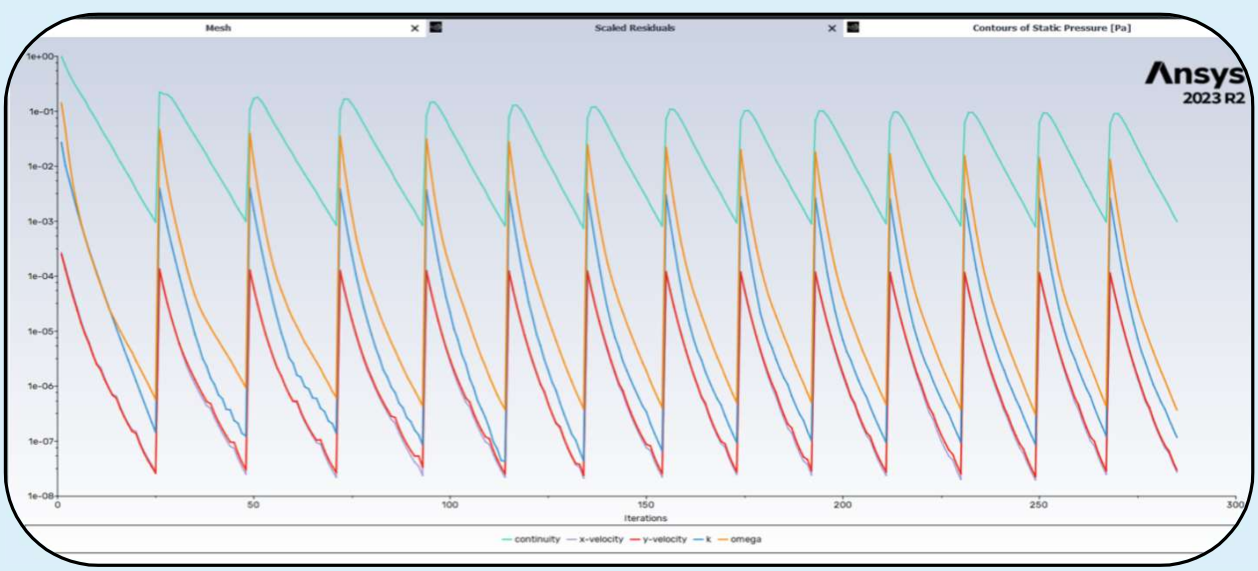


Features: The Project was simulated as a 2D unsteady incompressible flow model of a suction robot in ANSYS Fluent to analyse suction efficiency, pressure distribution, and airflow behaviour.

Key Outcomes:

- Optimised suction design for consistent airflow distribution.
- Identified low-pressure regions improving grasp stability.
- Provided CFD-based validation for robot end-effector performance.



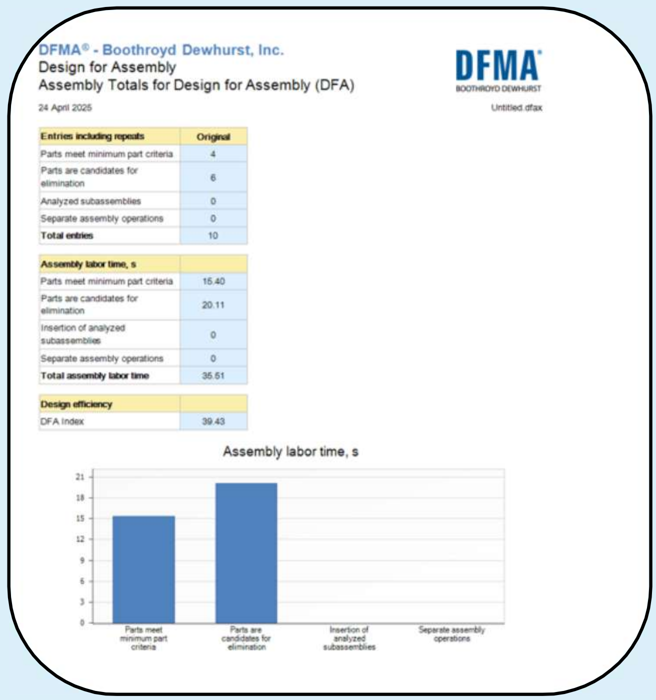


Project: *"Design and DFMA of a Prosthetic Leg"*

Features: The project was designed and integrated a prosthetic leg in **Autodesk Inventor** using DFMA principles to optimise strength, weight, and manufacturability, ensuring modular assembly and ease of maintenance.

Tools: Autodesk Inventor, DFMA principles

Material selection: Aluminium, Carbon fibre composite



DFMA® - Boothroyd Dewhurst, Inc.
Design for Assembly
Assembly Totals for Design for Assembly (DFA)

24 April 2025

Untitled.dfax

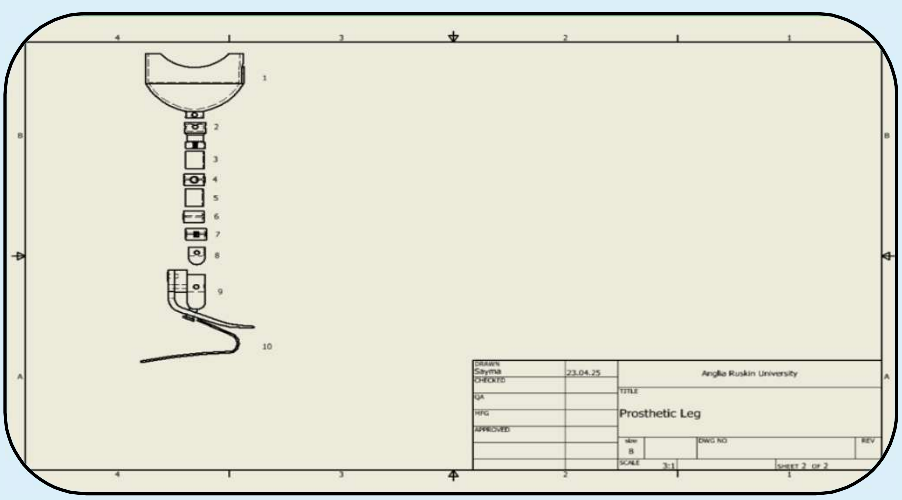
Entries including repeats	Original
Parts meet minimum part criteria	4
Parts are candidates for elimination	6
Analyzed subassemblies	0
Separate assembly operations	0
Total entries	10

Assembly labor time, s	
Parts meet minimum part criteria	15.40
Parts are candidates for elimination	20.11
Insertion of analyzed subassemblies	0
Separate assembly operations	0
Total assembly labor time	35.51

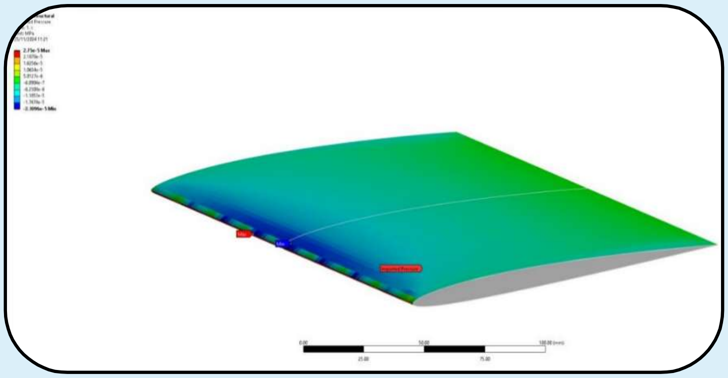
Design efficiency	
DFA Index	39.43

Outcomes

- Designed and **3D printed** a functional modular prosthetic leg prototype.
- Achieved a **lightweight and manufacturable** design suitable for scaled production.
- Improved **assembly efficiency** and part accessibility through DFMA principles.
- Demonstrated a **research-informed design approach**, translating benchmarking insights into practical engineering solutions.



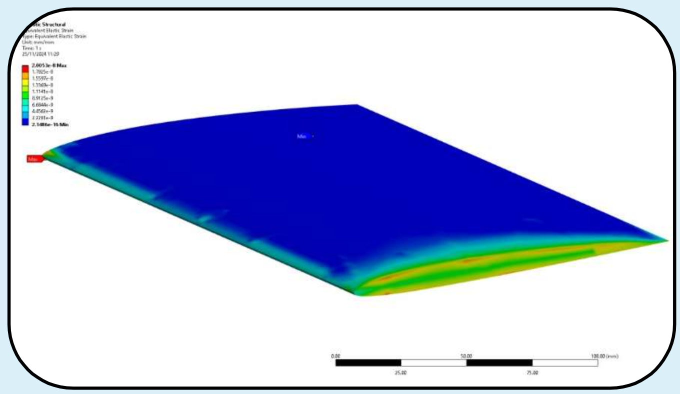
Project (Group Project): “*Aerovolt Small-Scale Wind Turbine Design*”

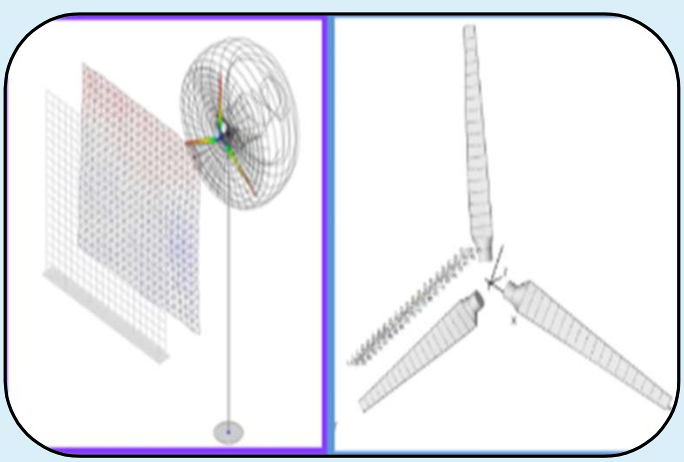


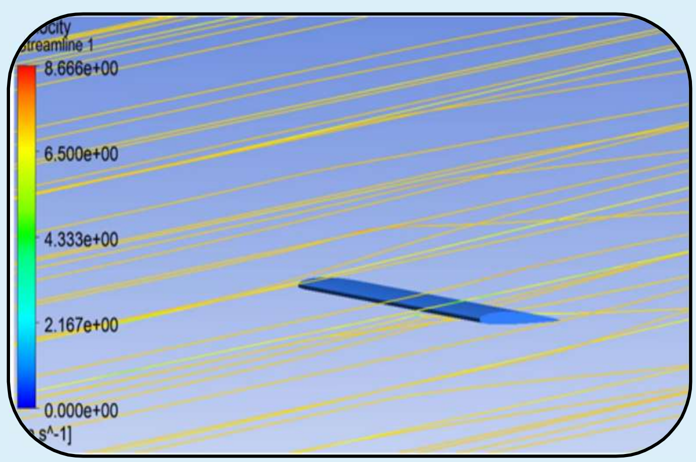
Features: This project was designed and analysed the small-scale wind turbine using **ANSYS** and **QBlade** to optimise blade aerodynamics, structural performance, and overall energy efficiency

Key Outcomes:

- Improved **lift-to-drag ratio** and **power coefficient (Cp)** through blade optimisation.
- Validated **structural integrity** and lightweight design using ANSYS.
- Demonstrated integration of **aerodynamic and structural analysis** for renewable energy applications.



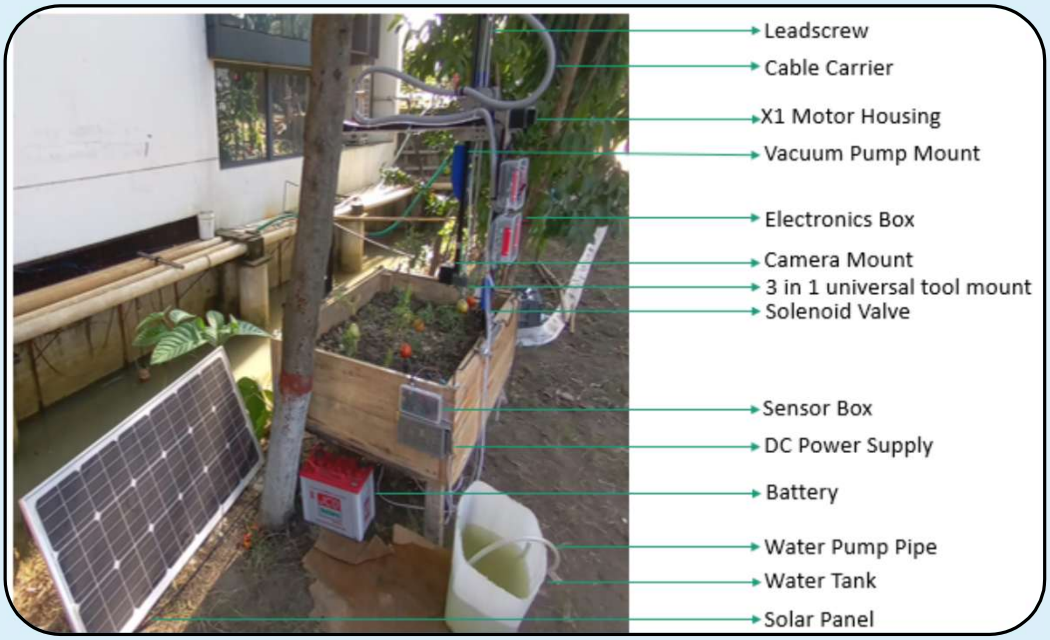




Thesis Research Project (Undergraduate)

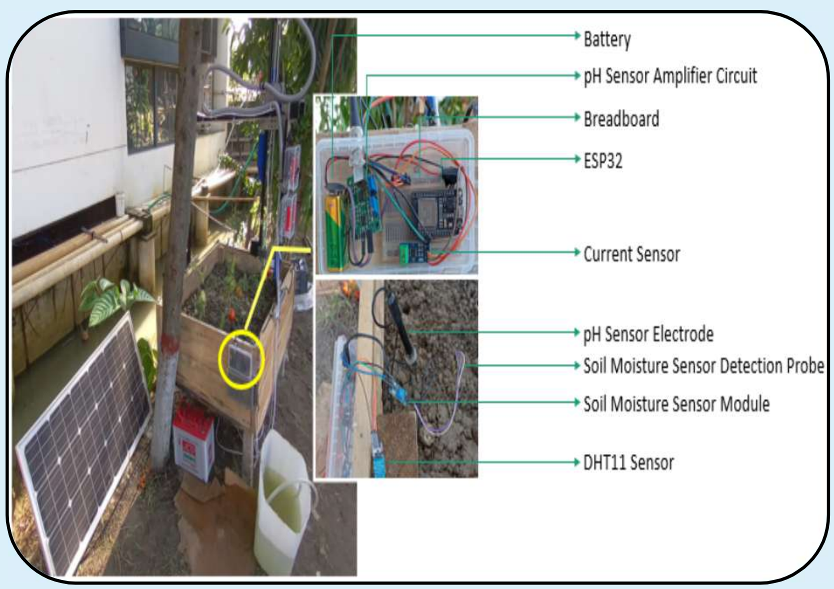
Thesis title: "*A Machine Learning-Based IoT-Enabled Farmbot for Smart Farming*"

Project Duration and Place: Jan 2023-Aug 2023



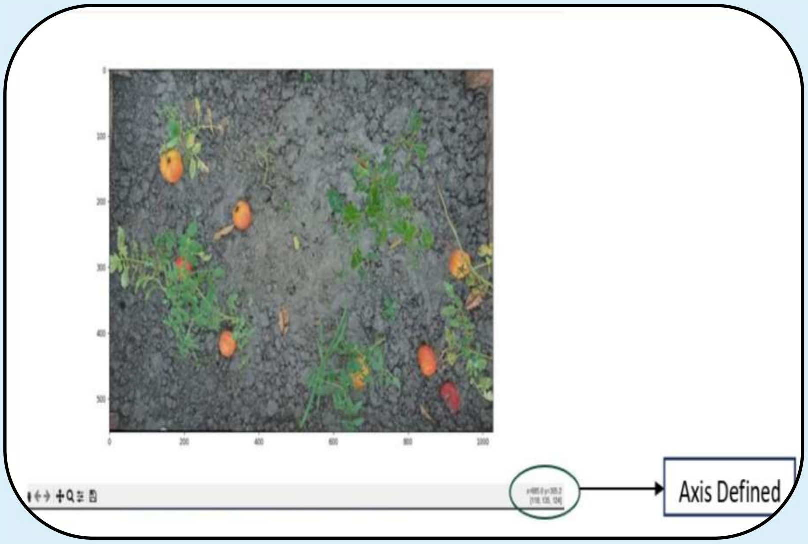
System Setup for Farmbot on Farming Bed

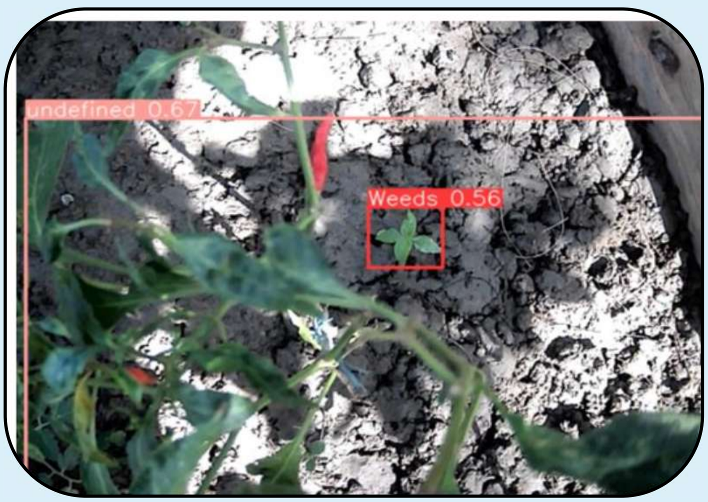
Description: In this project, a **Universal CNC Farmbot** was designed and developed to monitor tomato plants and automate farm operations. The system integrates a **CNC-based mechanism** for precise movement and control, along with a dedicated sensor box for real-time monitoring of plant conditions. This setup enables efficient crop management through automation, reducing manual effort and improving productivity.



Farmbot and Sensor Box Setup

These pictures include farming area mapping through image stitching using OpenCV, and the automated detection of weeds, as well as ripe, unripe, and defective tomatoes, using a YOLOv8 model. The detection model was trained on a custom dataset, created through the collection and preprocessing of tomato and weed images.





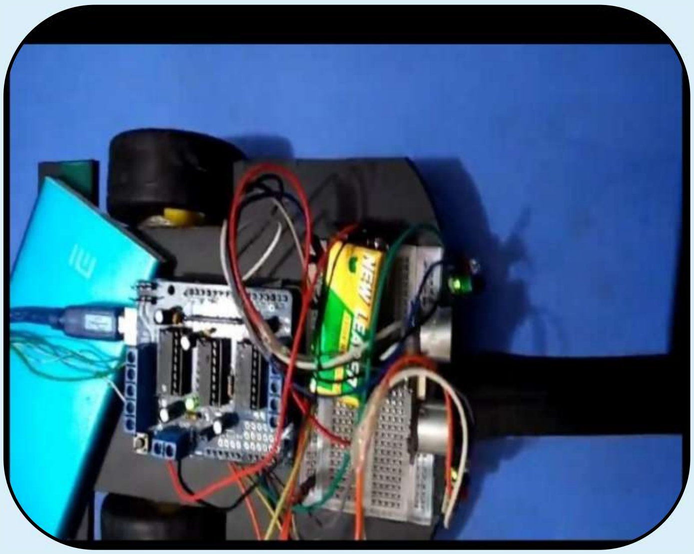
Outcomes

- Developed a **CNC Farmbot** for automated tomato farming.
- Enabled **real-time plant monitoring** and **weed/tomato detection** using YOLOv8.
- Improved **crop management efficiency** and reduced manual labor.

Weed and Tomato health detection

Additional Projects (Undergraduate)

Arduino-Based Line Following / Voice Controlled/Fire Sensor Robot using IR-Ultrasonic Sensor and Bluetooth module



At first, the project was started by making **line-following robots**. After that, it was upgraded to **a voice-controlled bot,** and afterwards, by adding some sensors, it was converted into a fire sensor bot.

Features:

Line following robot:

- Utilises **infrared sensors** to detect and follow lines on surfaces.
- Integrates **obstacle sensors** to prevent collisions and maintain uninterrupted line tracking.

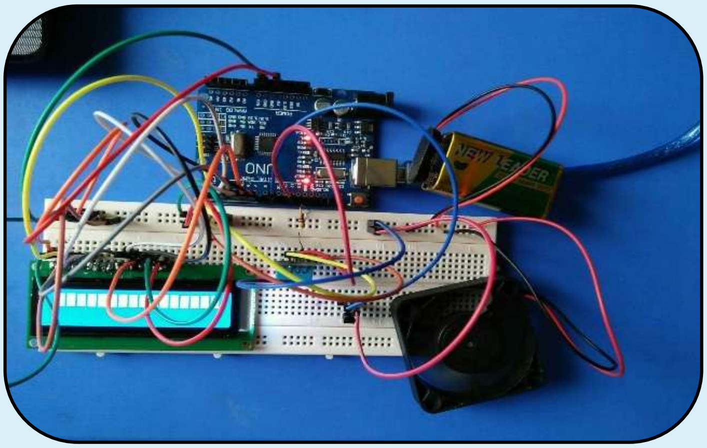
Voice-controlled bot features:

- Enables real-time bidirectional communication between the car and the mobile device.
- Utilises a dedicated mobile app installed on the user's smartphone to send voice commands to the car via Bluetooth.

Fire sensor bot features:

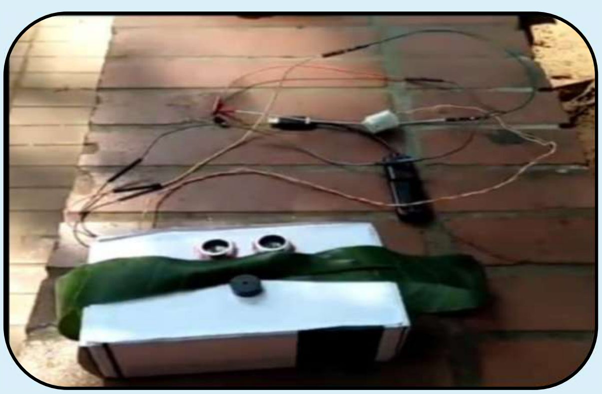
- Equipped with a fire sensor to detect flames or significant increases in temperature, enabling rapid response to potential fire hazards.
- Utilises IR sensors for obstacle avoidance, ensuring safe movement in dynamic environments while responding to fire threats.

A temperature-controlled fan system



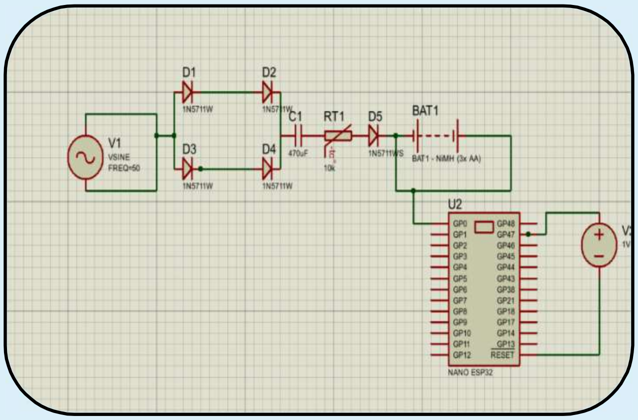
This project is a temperature-controlled fan. This temperature-controlled fan automatically adjusts its speed based on temperature, ensuring comfort and energy efficiency. It offers customizable temperature settings, quiet operation, and safety features, including overheat protection. LCDs or indicator lights provide real-time feedback, while multiple fan speeds offer additional control over airflow.

An Automated Blind Stick for Navigation and Obstacle Detection



Features: This project aimed to develop and simplify the circuit design of a blind stick that integrates obstacle detection sensors and a buzzer. It offers adjustable sensitivity and emergency features for enhanced safety and independence. Compact and lightweight, it ensures portability and comfort during daily use. This was designed in **Fritzing**, and **Arduino** has been used as a microcontroller.

Piezoelectric Phone Charger Harnessing Mechanical Energy for Mobile Power

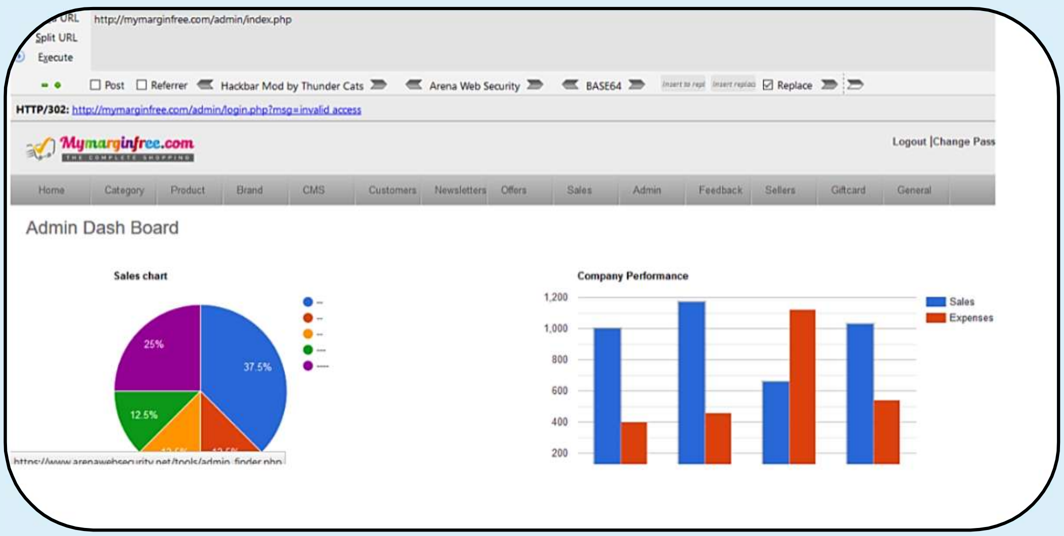


Features: The project was developed to simplify the circuit design in **Proteus** of a Piezoelectric phone charger.

This energy is then used to charge batteries, providing a sustainable and environmentally friendly power source. These chargers are portable, efficient, and can be integrated into various devices for on-the-go charging without the need for traditional power sources.

Ethical Hacking Projects

Session Hijacking (No redirect)



WAF Bypass

Publications

Selected Journal Papers:

- Zonayed, M., Tasnim, R., **Jhara, S. S.,** Mimona, M. A., Hussein, M. R., Mobarak, M. H., & Salma, U. (2025). Machine Learning and IoT in Healthcare: Recent Advancements, Challenges & Future Direction. *Advances in Biomarker Sciences and Technology*. **DOI**
- Tasnim, R., Zonayed, M., **Jhara, S. S.,** Hussein, M. R., Hossain, F., Saleheen, R. S., Islam, S. (in revision, will publish in 2025). An Ensemble Machine Learning Approach for the Prediction of Viral Infections using IoMT-Enabled Wearable Health Monitoring System. *Health System & Reform.*
- Zonayed, M., Hussein, M. R., Tasnim, R., **Jhara, S. S.,** Tasnim, Z., Hasan, M., Hossain, F. (in revision, 2025). Advancing Precision Agriculture: A Comprehensive Review of Machine Learning Applications with Limitations & Future Prospects. *Computer and Electronics in Agriculture*
- Shaikat, A. S., Hussein, M. R., Tasnim, R., Zonayed, M., **Jhara, S. S.,** Rahman, M. M., & Mokhter, A. H. (2024). An IoT-enabled Artefact Protection System for a Museum using Computer Vision. *Journal of Engineering Advancements*, *5*(04), 123-131. **DOI**

Selected Conference Papers:

- Hussein, M. R., Shaikat, A. S., Tasnim, R., Zonayed, M., Amin, A. A., Rahman, S., **Jhara S. S.,** "An Automated Floor Cleaner Robot for Domestic and Industrial Applications," *2024 International Conference on Decision Aid Sciences and Applications (DASA)*, Manama, Bahrain, 2024, pp. 1-7, *IEEE.* **DOI**

Book Chapter

- Nusrat Zahan Ramesha, Md Sadatuzzama Saagoto, Md Zonayed, **Sayma Sultana Jhara**, Rumana Tasnim, Enamul Huq. **Chapter 2- Sensors & Actuators**. In M. M. Rahman, F. Mahbub, R. Tasnim, & R. U. Saleheen (Eds.), Mechatronics: Fundamentals and Applications. Springer Singapore (2024) (In Press) (Hardcover ISBN- 978-981-97-7116-5). **DOI**

Extracurricular Activities

Industry Visit/Workshop

Eaton Industries· 1-day visit · Hands-on exposure to automation systems · July 2025 · Bonn, Germany

Medical Equipment Training Session (with Jennifer PE) · Practical training on equipment operation, safety, and common errors · Dec 2024

Ford Dunton Technical Centre · Gained knowledge of 3D printing, robotic control systems, and powertrain simulation · Nov 2024, UK

1-day visit to **IKA** , RWTH Aachen University, Aachen, Germany (July 2025)



Work Shadow

Work Shadow, London · **Cubic Transport** ·Supervisor: Lalitha Ande (Sr. System Engineer) · 3 days (July 2025) · London, UK


Networking

Summer School · **RWTH Aachen University** ·July 2025 · Aachen, Germany

Networking Event · **British Council Scholars**, 100 scholars from 25 UK universities · **Glasgow University**, Scotland · Mar 2025

Project Showcasing and Debate Competition

Winner of the Inter-Departmental Project Competition at **Mechatronics Tech Fest**, WUB · 2023(Published in Some National Papers of Bangladesh)

Competitor in the project showcasing at Ignition **National Mechanical Fest**, KUET · 2023

Winner of the Inter-Departmental Debate Competition at Mechatronics Tech Fest, WUB · 2018



Club and Other Activities

Assistant Community Manager at Mechatronics Club, WUB · Jan 2018- Jan 2020

Rover in **Bangladesh Scout**. Air Region (2013-2015)

Campus Coordinator ·Membership ID:0201 · BYOSD and Career care · Jan 2020 - Dec 2020

Leadership Experiences

Vice-President, Bangladesh Society for Total Quality Management (Student Chapter)

(Jan 2018 - Apr 2019)

BSTQM is a voluntary organisation promoting Total Quality Management (TQM) in Bangladesh. The BSTQM Student Chapter includes the enhancement of TQM culture among the students.

Stream Manager 21st International Convention on SQCC, **BSTQM** (Student chapter) · 2018

Stream Manager 10th National Annual Convention on Education, **BSTQM** (student chapter) · 2018

Participant in the 12th International Convention on SQCC, CMS(India) as a team leader of SQCC from New Dhaka International School (Jr. Group) · 2009

Recommendation

Rumana Tasnim

(Undergraduate Thesis Supervisor)

Assistant Professor

Department of Mechatronics Engineering

World University of Bangladesh

Email: rumana@mte.wub.edu.bd

Dr. Alireza Sanaei

(Master's Dissertation Supervisor)

Senior Lecturer

Electronics & Robotics Engineering

Anglia Ruskin University

Email: as61@aru.ac.uk